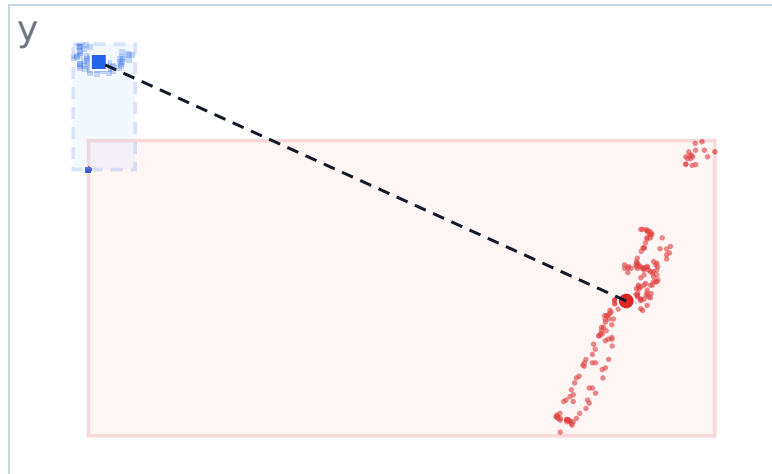


(a) High-confidence relation

● subject ■ object



heater → close by → trash can

Open3DSG: $Z = 0.853$; rank 19

XY distance = 4.33 m

(b) Predicate-geometry compatibility

Predicate T

close by / proximity

Pair measurements G

distance, height, overlap

Compatibility $C_{tr}(T,G)$

linked counterfactual ordering
transformation averaging

$C_{tr} = 0.003$

Compatibility inputs: T and G

Source relation score Z

used only in re-ranking

$S = Z \times C_{tr}(T,G)$

within applicable family

(c) Re-ranking

19

source rank

178

re-ranked

Family-aware re-ranking

proximity / vertical: re-ranked
support/contact: source order

Joint evaluation

Exact-label Recall@K
Verifier-derived Violation@K